**EXPERIMENT – 4** **DATE: 22-10-2026**

**INDUSTRIAL HVAC CONTROL FIRMWARE: ADC TEMPERATURE ACQUISITION AND TIMER-INTERRUPT PWM FAN CONTROL**

**Objectives of this Experiment:**

1. To understand the operation of the Arduino UNO ADC for analog temperature acquisition.
2. To interface an LM35 temperature sensor with the Arduino UNO.
3. To acquire and convert the LM35 analog output into temperature values using Embedded C programming.
4. To implement timer-interrupt-based software PWM for fan-speed control.
5. To control an HVAC fan according to the measured temperature.
6. To display the temperature and corresponding PWM duty cycle through the Proteus Virtual Terminal.
7. To simulate and verify the HVAC control firmware using Proteus software.

**OUTPUT PARAMETER TABLE – VALUES TO BE RECORDED**

| S. No. | Output Parameter | Expected Value / Condition | Practical / Simulated Observation |
|---|---|---|---|
| **1** | **Temperature Acquisition** | LM35 read through A0 | Verified |
| **2** | **Temperature < 25°C** | PWM = 0% | Verified |
| **3** | **Temperature 25–29°C** | PWM = 30% | Verified |
| **4** | **Temperature 30–34°C** | PWM = 60% | Verified |
| **5** | **Temperature ≥ 35°C** | PWM = 90% | Verified |
| **6** | **Fan Control** | Fan responds to PWM | Verified |
| **7** | **Serial Output** | Temp. and PWM displayed | Verified |

**RUBRIC:**

| Objective | Assessment Criterion | Weightage | Marks |
|---|---|---|---|
| Develop the Proteus circuit | Circuit Design and Configuration | 2.5 | |
| Develop the Embedded C program | Program Development | 2.5 | |
| Implement sensor processing | Sensor Acquisition and Decision Logic | 2.5 | |
| Verify system operation | Simulation and Result Analysis | 2.5 | |
| **Total** | | **10** | |

Experiment uploaded in GitHub on :

https://github.com/madurivadapalli-tech/192311368.EMBEDDED/blob/main/LAB%20EXPERIMENTS/EXPERIMENT%204.11368.pdf